

Chapter 6: Bases and dimension

In this part of the module, we develop the theory of bases for a finitely generated vector space V further. We will see that

- every collection of vectors generating (=spanning) the space V contains a basis for V (basis selection algorithm);
- every finitely generated vector space has a basis;
- given any collection of linearly independent vectors in V , there exists a basis for V containing the prescribed vectors (basis extension algorithm).

We will prove that *any two bases for V have the exact same number of elements* (invariant basis number property); this number is called the *dimension* $\dim(V)$ of V . Knowing the dimension of V simplifies finding a basis since we know in advance how many vectors we need.

Section 6.1: The basis selection algorithm

1 The basis selection algorithm

Suppose that V is a finitely generated vector space, and suppose that the (finitely many) vectors $w_1, w_2, \dots, w_n$ generate V . Label these vectors "good" or "bad" as follows, for $k = 1, 2, \dots, n$:

- w_k is "good" if it is **not** in the span of the vectors that have been labelled good so far;
- w_k is "bad" if it **is** in the span of the good vectors so far.

When all the vectors are labelled, the good vectors among them are a basis for V .

Recall that $\text{span}(\emptyset) = \{\underline{0}\}$, so *the first non-zero vector among the w_j is automatically good*.

2 Proof: basis selection algorithm produces a basis

If the same vector is listed several times, only its first occurrence can possibly be labelled "good". So **we may assume** from the outset **that $w_i \neq w_j$ for $i \neq j$** .

Write $S = \{w_1, w_2, \dots, w_n\}$. By hypothesis, $\text{span}(S) = V$.

Run the algorithm on the list of vectors $w_1, w_2, \dots, w_n$ so we know which ones are "bad". Let $T \subseteq S$ be the set of bad vectors in S . Then $S \setminus T$ is the set of good vectors.

We show first that **the good vectors generate V** :

By construction, every bad vector is a linear combination of good vectors, so $T \subseteq \text{span}(S \setminus T)$. Then

$$\text{span}(S \setminus T) = \text{span}(S) = V$$

by "omitting redundant generators" (Chapter 3, slide 36).

In words: the good vectors generate V , just as claimed.

3 Verification of basis selection algorithm (continued)

Notation: let g_1 be the first good vector the algorithm produces, g_2 the second good vector and so on, with g_k being the last good vector, so that

$$S \setminus T = \{g_1, g_2, \dots, g_k\}.$$

Still need to check: **$g_1, g_2, \dots, g_k$ are linearly independent**

- $k = 0$: This means $S \setminus T = \emptyset$. Then all given vectors are $\underline{0}$, so $S = \{\underline{0}\}$. But $V = \text{span}(S) = \{\underline{0}\}$ in this case, and $S \setminus T = \emptyset$ is indeed a basis for V .

- $k = 1$: Since a single *non-zero* vector is linearly independent, we know that g_1 is a basis for V .

So we still need to deal with the case $k \geq 2$.

4 Verification of basis selection algorithm (continued)

Assume towards a contradiction that $g_1, g_2, \dots, g_k$ are linearly dependent. Then we find scalars λ_j , **not all zero**, with

$$\lambda_1 g_1 + \lambda_2 g_2 + \dots + \lambda_k g_k = \underline{0}.$$

Let m be the largest index with $\lambda_m \neq 0$ so that in fact

$$\lambda_1 g_1 + \lambda_2 g_2 + \dots + \lambda_m g_m = \underline{0}.$$

If $m = 1$ then (4) says $\lambda_1 g_1 = \underline{0}$ with $\lambda_1 \neq 0$, so $g_1 = \underline{0}$. But good vectors cannot be zero vectors. So **we must have $m > 1$** . Thus by re-arranging (4) we obtain

$$g_m = -\frac{\lambda_1}{\lambda_m} g_1 - \frac{\lambda_2}{\lambda_m} g_2 - \dots - \frac{\lambda_{m-1}}{\lambda_m} g_{m-1},$$

that is, g_m is in the span of the previously found good vectors. But then g_m must have been labelled "bad" – contradiction! ■

5 Basis selection algorithm: Example

Consider the following vectors in $\mathbb{R}^3$:

$$w_1 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, w_2 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, w_3 = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, w_4 = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}, w_5 = \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix}$$

These vectors generate $\mathbb{R}^3$. Verifying this amounts to showing that the vector equation

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda_1 w_1 + \lambda_2 w_2 + \lambda_3 w_3 + \lambda_4 w_4 + \lambda_5 w_5$$

always has a solution (the variables are the λ_j). This amounts to solving a system of three linear equations in five variables, a routine task. (Do it as an exercise!)

6 Basis selection algorithm: Example (continued)

By basis selection algorithm we know that the list

$$w_1, w_2, w_3, w_4, w_5$$

contains a basis of $\mathbb{R}^3$:

- $w_1 \neq \underline{0}$, so w_1 is "good"
- $w_2 = \underline{0} = 0w_1 \in \text{span}\{w_1\}$, so w_2 is "bad"
- $w_3 \notin \text{span}\{w_1\}$ (as w_3 is not a multiple of w_1), so w_3 is "good"
- $w_4 = -w_1 + w_3 \in \text{span}\{w_1, w_3\}$, so w_4 is "bad"
- $w_5 \notin \text{span}\{w_1, w_3\}$ as $w_5 = \lambda w_1 + \mu w_3$ has no solution (this amounts to solving a system of three linear equations with two variables; do it as an exercise!); so w_5 is "good"

The basis selection algorithm terminates now; the resulting basis of $\mathbb{R}^3$ is w_1, w_3, w_5 .

7 Every finitely generated vector space has a basis

The basis selection algorithm gives the following crucial result:

Every finitely generated vector space has a basis.

Proof. V finitely generated vector space (cf. slide 31 of Chapter 3)

$\implies$ find vectors $w_1, w_2, \dots, w_n \in V$ which generate V

By basis selection algorithm, we can find a basis for V consisting of some (but maybe not all) of the vectors $w_1, w_2, \dots, w_n$.

In particular: V has a basis (since we've just found one). ■

8 **Aside: Does *every* vector space have a basis?**

So what about a vector spaces that isn't **not** finitely generated: does it have a basis?

It turns out that this question does not have a straightforward answer. The answer depends on whether you believe in the "axiom of choice" or not! In other words, the answer depends on the "set theory" you work with.

In most of mainstream mathematics, the "axiom of choice" is accepted to hold (and it can be shown that this does not create any new logical problems). With that, it is true that **every vector space has a basis** though we will not discuss the proof in general.

Section 6.2: The basis extension algorithm

9 The basis extension algorithm

Basis extension algorithm

Suppose the $\mathbb{R}$ -vector space V is generated by the vectors $v_1, v_2, \dots, v_n$. Suppose further that $w_1, w_2, \dots, w_k$ is a (possibly empty) list of linearly independent vectors in V .

Running the basis selection algorithm on the list of vectors

$$w_1, w_2, \dots, w_k, v_1, v_2, \dots, v_n$$

(in this order!) produces a basis for V which contains all the given vectors w_j .

In other words, we have "extended" the list $w_1, \dots, w_k$ to a basis for V .

10 The basis extension algorithm (proof)

Proof. Since $v_1, v_2, \dots, v_n$ generate V so do the vectors

$$w_1, w_2, \dots, w_k, v_1, v_2, \dots, v_n ;$$

indeed, we have

$$V = \text{span} \{v_1, v_2, \dots, v_n\} \subseteq \text{span} \{w_1, \dots, w_k, v_1, \dots, v_n\} \subseteq V$$

(using the monotonicity of span for the first inclusion, see slide 19 of chapter 3).

Now use the basis selection algorithm (slide 1 of this chapter) to find a basis within the list of vectors $w_1, \dots, w_k, v_1, \dots, v_n$.

Since $w_1, w_2, \dots, w_k$ are linearly independent, no w_i is in the span of the other w_j (see slide 15 of chapter 4), so **all of the w_j are marked "good"** and are thus elements of the basis produced by the basis selection algorithm. ■

11 Basis extension algorithm: Example

The two polynomials

$$p_1 = 2x^3 + x^2 - x \quad \text{and} \quad p_2 = x^3 + x^2 - x \quad \text{in } \mathcal{P}_3$$

are linearly independent (check this!). Let's try to extend the list p_1, p_2 to a basis of $\mathcal{P}_3$.

We know that $x^3, x^2, x, 1$ generate $\mathcal{P}_3$. We can now apply the basis extension algorithm — this amounts to running the basis selection algorithm on the list $p_1, p_2, x^3, x^2, x, 1$:

- $p_1 \neq \underline{0}$, so p_1 is labelled "good"
- $p_2 \notin \text{span} \{p_1\}$ because p_2 is not a scalar multiple of p_1 , so p_2 is labelled "good"

Of course **we knew this already** — the list of vectors we're extending is linearly independent, hence all its vectors will be labelled "good".

12 Basis extension algorithm: Example (continued)

- $x^3 = (2x^3 + x^2 - x) - (x^3 + x^2 - x) = p_1 - p_2 \in \text{span}\{p_1, p_2\}$, so x^3 is labelled "bad"
- $x^2 \notin \text{span}\{p_1, p_2\}$, so x^2 is labelled "good" — why exactly? Try to find $\lambda, \mu \in \mathbb{R}$ such that

$$x^2 = \lambda p_1 + \mu p_2 = (2\lambda + \mu)x^3 + (\lambda + \mu)x^2 - (\lambda + \mu)x.$$

Such λ, μ are precisely the solutions of the following system:

$$\left. \begin{array}{ll} \text{(coefficient of } x^3) & 2\lambda + \mu = 0 \\ \text{(coefficient of } x^2) & \lambda + \mu = 1 \\ \text{(coefficient of } x) & -\lambda - \mu = 0 \end{array} \right\}$$

There is no solution as adding the second to the third equation results in the equation $0 = 1$.

13 Basis extension algorithm: Example (continued)

- $x = (2x^3 + x^2 - x) - 2(x^3 + x^2 - x) + x^2 = p_1 - 2p_2 + x^2 \in \text{span}\{p_1, p_2, x^2\}$, so x is labelled "bad"
- $1 \notin \text{span}\{p_1, p_2, x^2\}$ (check this!), so 1 is labelled "good"

So the list $p_1, p_2, x^2, 1$ is a basis for $\mathcal{P}_3$ extending the list p_1, p_2 .

We can get different answers by listing the vectors in a different order: running basis selection algorithms on the list

$$p_1, p_2, 1, x, x^2, x^3$$

results in the basis $p_1, p_2, 1, x$ which also extends the list p_1, p_2 .

Observe that **both bases have the same number of elements!** This is no coincidence, as we'll see soon.

Section 6.3: The invariant basis number property

The invariant basis number property asserts that *any two bases of a vector space have the same number of elements*.

14 Homogeneous systems of linear equations revisited

The "general" homogeneous system of linear equations has this form:

$$\left. \begin{array}{cccccc} a_{1,1}x_1 & + & a_{1,2}x_2 & + & \dots & + & a_{1,n}x_n & = & 0 \\ a_{2,1}x_1 & + & a_{2,2}x_2 & + & \dots & + & a_{2,n}x_n & = & 0 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ a_{m,1}x_1 & + & a_{m,2}x_2 & + & \dots & + & a_{m,n}x_n & = & 0 \end{array} \right\} \quad (\text{S})$$

This system is homogeneous, so it must have a solution, *viz.*, the trivial solution (all $x_j = 0$). But *if the number of variables exceeds the number of equations, we are guaranteed to also have a [non-trivial](#) solution*:

If $n > m$ the system (S) has a **non-trivial** solution, *i.e.*, a solution with at least one $x_j \neq 0$.

15 Homogeneous systems of linear equations revisited (continued)

Proof. We use GAUSS elimination to transform (S) into a system in row echelon form. In the resulting system, at most m variables can occur as the left-most variable in some equation; that means that at least $n - m > 0$ variables do *not* occur as leftmost variable somewhere, hence [at least \$n - m > 0\$ variables remain as parameters in the solution](#).

Choosing an arbitrary non-zero value for one of the parameters (and arbitrary values for the others, for example, zero) yields a non-trivial solution. ■

16 More than m linear combinations of m vectors are linearly dependent

If you start with "a few" vectors $v_1, \dots, v_m$ and form "too many" linear combinations $w_1, \dots, w_n$ from them, then the vectors $w_1, \dots, w_n$ must be linearly dependent. The meaning of "too many" is precisely that n exceeds m . Here is the formal statement:

- V an $\mathbb{R}$ -vector space
- $v_1, v_2, \dots, v_m \in V$ arbitrary vectors
- $w_1, w_2, \dots, w_n \in \text{span}\{v_1, v_2, \dots, v_m\}$

If $n > m$ then the vectors $w_1, w_2, \dots, w_n$ are linearly dependent.

Proof. Suppose $n > m$. We want to show: there are scalars λ_j , **not all zero**, such that

$$\lambda_1 w_1 + \lambda_2 w_2 + \dots + \lambda_n w_n = \underline{0} .$$

17 More than m linear combinations of m vectors are linearly dependent (continued)

Since $w_j \in \text{span}\{v_1, v_2, \dots, v_m\}$ we find scalars $\mu_{i,j}$ such that

$$w_j = \mu_{1,j}v_1 + \mu_{2,j}v_2 + \dots + \mu_{m,j}v_m \quad \text{for } 1 \leq j \leq n .$$

Rewriting $\lambda_1 w_1 + \lambda_2 w_2 + \dots + \lambda_n w_n$ in these terms gives:

$$\begin{aligned} \sum_{j=1}^n \lambda_j w_j &= \sum_{j=1}^n \lambda_j (\mu_{1,j}v_1 + \mu_{2,j}v_2 + \dots + \mu_{m,j}v_m) \\ &= \sum_{j=1}^n \lambda_j \sum_{i=1}^m \mu_{i,j}v_i = \sum_{i=1}^m \left(\sum_{j=1}^n \mu_{i,j} \lambda_j \right) \cdot v_i \end{aligned}$$

We want this to be the zero vector $\underline{o}$. It is **enough** (but not necessary!) that the coefficients of the v_i (in parentheses) vanish:

$$\sum_{j=1}^n \mu_{i,j} \lambda_j = 0 \quad \text{for } 1 \leq i \leq m$$

18 More than m linear combinations of m vectors are linearly dependent (continued)

This means precisely that the λ_j are a solution of the following homogeneous system of linear equations:

$$\left. \begin{array}{ccccccc} \mu_{1,1}\lambda_1 & + & \mu_{1,2}\lambda_2 & + & \dots & + & \mu_{1,n}\lambda_n & = & 0 \\ \mu_{2,1}\lambda_1 & + & \mu_{2,2}\lambda_2 & + & \dots & + & \mu_{2,n}\lambda_n & = & 0 \\ \vdots & & \vdots & & \ddots & & \vdots & = & \vdots \\ \mu_{m,1}\lambda_1 & + & \mu_{m,2}\lambda_2 & + & \dots & + & \mu_{m,n}\lambda_n & = & 0 \end{array} \right\} \quad (\text{S})$$

By hypothesis $n > m$, so from slide 14 of this chapter we know that **there exists a non-trivial solution**. This means we find scalars λ_j , not all zero, such that

$$\lambda_1 w_1 + \lambda_2 w_2 + \dots + \lambda_n w_n = \underline{o}$$

proving that $w_1, w_2, \dots, w_n$ are linearly dependent. ■

19 Invariant basis number property

The number of elements of a basis is the same in any basis

Let V be an $\mathbb{R}$ -vector space. We will show that the number of elements in any basis of V is the same.

This is sometimes expressed by saying that "vector spaces have the invariant basis number property".

Invariant Basis Number Property

- V an $\mathbb{R}$ -vector space

- $b_1, b_2, \dots, b_m$ a basis of V
- $w_1, w_2, \dots, w_n$ another basis of V

Then we must have $m = n$. That is, if V has **some** basis consisting of m elements, then **every** basis for V has precisely m elements.

20 Invariant basis number property (continued)

Proof. The case $m = 0$ is trivial: $V = \{\underline{0}\}$ in this case, and $\emptyset$ is the only basis of V . So there is nothing to show.

We need to deal with the case $m > 0$.

We show first that n *cannot exceed* m . Reason: Assume that $n > m$. We know

$$w_j \in V = \text{span}\{b_1, \dots, b_m\} \quad (1 \leq j \leq n),$$

and from slide 16 of this chapter we know that $w_1, \dots, w_n$ must be linearly dependent (recall $n > m$), hence they cannot form a basis for V . Contradiction!

So we know that $n \leq m$ must hold.

We now repeat the argument, with the roles of the bases $b_1, b_2, \dots, b_m$ and $w_1, w_2, \dots, w_n$ reversed, to conclude that m cannot exceed n either. This proves $m = n$. ■

Section 6.4: The dimension of a vector space

21 The dimension of a finitely generated vector space

Definition. Let V be a finitely generated vector space. The **dimension** of V is the number of vectors in any basis for V . We denote the dimension of V

by the symbol $\dim(V)$.

Note that "dimension" refers to vector spaces, not to individual vectors, and not to arbitrary sets!

Remark: It follows from the invariant basis number property (slide 19 of this chapter) that the dimension of a vector space is independent of the particular basis that is used to calculate it (as every basis has the same number of elements). In other words:

The invariant basis number property guarantees that the dimension of a finitely generated vector space is well-defined.

22 Terminology: Finite-dimensional vector spaces

An $\mathbb{R}$ -vector space V is called *finite-dimensional* if it has a basis consisting of finitely many vectors.

The vector space V is finite-dimensional if and only if V is finitely generated; we write $\dim(V) < \infty$ in this case.

In case V is not finite-dimensional, we say that V is *infinite-dimensional* and write $\dim(V) = \infty$.

Specific examples of infinite-dimensional vector spaces are:

- The space $\mathcal{S}$ of sequences of real numbers
- The space $\mathcal{P}$ of all polynomials with real coefficients

23 List of known dimensions

Here is an overview of the dimensions we know already:

- The trivial vector space $\{\underline{0}\}$ has dimension 0. The *only* basis of this vector space is the empty basis $\emptyset$.

- The vector space $\mathbb{R}^n$ has dimension n , the unit vectors $e_1, e_2, \dots, e_n$ form a basis (slide 2 of chapter 5).
- The vector space $\mathcal{P}_n$ of polynomials of degree at most n has dimension $n + 1$. The $n + 1$ monomials $x^n, x^{n-1}, \dots, x, 1$ form a basis (slide 3 of chapter 5).
- The vector space $\mathcal{M}_{m,n}$ of m -by- n matrices with real entries has dimension mn . The mn matrices with a single entry 1, and all other entries 0, form a basis (slide 4 of chapter 5).

24 Recognising a basis

Let V be an n -dimensional $\mathbb{R}$ -vector space, and let $v_1, v_2, \dots, v_n \in V$.

- (i) If $v_1, v_2, \dots, v_n$ generate V , they are linearly independent (and thus are a basis for V).
- (ii) If $v_1, v_2, \dots, v_n$ are linearly independent, they generate V (and thus are a basis for V).

In words: to check that n vectors form a basis, with $n = \dim(V)$, it is enough to check that they generate V , **or** to check that they are linearly independent. The other condition comes for free! (Well, almost for free: we need to know the dimension of V in advance.)

25 Recognising a basis (continued)

Proof. (i) By the basis selection algorithm (slide 1 of this chapter) we can find a basis consisting of *some* of the vectors v_j . But this basis must have n elements, as $n = \dim(V)$, see slide 19 of this chapter, so in fact the basis must consist of *all* of the v_j .

That is, $v_1, v_2, \dots, v_n$ is already a basis; hence the vector are linearly independent.

(ii) Follows similarly using basis extension algorithm (left as an exercise). ■

26 Recognising non-bases

Let V be an m -dimensional vector space.

- (i) Fewer than m vectors in V cannot generate V .
- (ii) Any set of more than m vectors in V is linearly dependent.

Proof. • (i) Left as an exercise.

- (ii) It is enough to show that n vectors cannot be linearly independent if $n > m$. Indeed, as $\dim V = m$ there is a basis with m elements. This basis generates the vector space V , so each of the given n vectors is a linear combination of the m basis vectors. Consequently the given n vectors are linearly dependent, see slide 16 of this chapter. ■

27 A shortcut for the basis selection algorithm

The results on the previous slide give a potential shortcut for the basis selection algorithm for vector spaces with known dimension:

Basis selection algorithm for m -dimensional vector space

Let V be an m -dimensional vector space, and suppose that the vectors $w_1, w_2, \dots, w_n$ generate V . Label the vectors "good" or "bad" in turn *until m good vectors have been found*:

Supposing we have labelled $w_1, \dots, w_{k-1}$ already for $k \leq n$, we say w_k is

- "good" if it is **not** in the span of the vectors that have been labelled good so far;
- "bad" if it **is** in the span of the good vectors.

When m good vectors have been found, these m vectors form a basis for V . *All remaining unlabelled w_j (if any are left over) are bad.*

Section 6.5: Subspaces of finitely generated vector spaces

In this section, we will address the following questions:

- Is the subspace of a finitely generated vector space itself finitely generated? (Yes!)
- Can the dimension of the subspace be larger than the dimension of the ambient vector space? (No!)

28 Dimension and subspaces

Let V be a finitely generated $\mathbb{R}$ -vector space, and let $U \subseteq V$ be a subspace of V . Then:

- U is finitely generated
- $\dim(U) \leq \dim(V)$
- $\dim(U) = \dim(V)$ if and only if $U = V$

Proof. We remark first that if $U = \{\underline{0}\}$ we have $\dim(U) = 0$, and there is nothing to show.

So suppose from now on, for the remainder of the proof, that $U \neq \{\underline{0}\}$. Our strategy is to "construct" a basis for U by choosing one vector at a time, and proving that this process must necessarily stop after at most $\dim(V)$ steps.

29 Dimension and subspaces (ct'd)

We choose vectors $u_1, u_2, \dots \in U$, as follows:

- Choose a vector $u_1 \in U$, $u_1 \neq \underline{0}$. We can do that since we are considering the case $U \neq \{\underline{0}\}$. Note that u_1 is linearly independent.

- Suppose we already have a list of linearly independent vectors

$$u_1, u_2, \dots, u_k \in U.$$

Two things can happen:

- The list is maximal (we cannot add another vectors from U and keep linear independence), and thus a basis for U (slide 16 of Chapter 5). Continue with next slide.
- Otherwise, we find a vector $u_{k+1} \in U$ such that $u_1, u_2, \dots, u_k, u_{k+1}$ are linearly independent. Choose such a u_{k+1} , and repeat.

This construction process must terminate with finitely many vectors. The reason is documented on the next slide.

30 Dimension and subspaces (continued)

The construction process above must terminate with finitely many vectors. For the vectors u_j are linearly independent in V (as they are linearly independent in U); but $\dim(V) < \infty$, and there cannot be more than $\dim(V)$ linearly independent vectors in V ! (See slide 26 of this chapter.)

In other words, **because V is finite-dimensional** we reach a number k such that

$$u_1, u_2, \dots, u_k$$

is a maximal list of linearly independent vectors in U , with $k \leq \dim(V)$.

The vectors $u_1, u_2, \dots, u_k$ are automatically a basis for U , by slide 16 of Chapter 5, so that $\dim(U) = k \leq \dim(V)$. — If $k = \dim(V)$ the vectors $u_1, u_2, \dots, u_k$ also form a basis for V (slide 24 of this chapter) so that U and V have the same basis, hence are equal: $U = V$. ■

31 Bases and subspaces

It is sometimes useful to be aware of the following fact:

Let V be a finite dimensional vector space, and let $U \subseteq V$ be a subspace. Then U is finite dimensional, and any basis of U can be extended to a basis of V .

Proof. We already know $\dim(U) \leq \dim(V)$. Let $u_1, u_2, \dots, u_k$ be a basis for U . These vectors remain linearly independent when considered as elements of V . By basis extension theorem, we can extend this list to a basis for V . ■

32 Summary

- Every finite collection of vectors generating a vector space V contains a basis for V (basis selection algorithm).
- Every finitely generated vector space has a basis.
- Any finite collection of linearly independent vectors in the finitely generated vector space V can be extended to a basis for V (basis extension algorithm).
- Let V be a finitely generated vector space. Any two bases for V have the same number of elements (invariant basis number property). This number is called the *dimension* of V , and is denoted $\dim(V)$.
- The dimension of a vector space equals the maximal size of a collection of linearly independent vectors in V , and also the minimal size of a collection of vectors generating V .
- A subspace U of a finitely generated vector space V is itself finitely generated, and $\dim(U) \leq \dim(V)$.